Two matter particles in the local dS_3 model: physical norms and an uncrossed limit

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Abstract

We derive the physical norm of the scalar observer channel reached by two ordered creations of distinguishable matter species. The full tensor product contains spinning channels; these are not discarded in completeness. A proper-mass half-period shift gives bounded creation blocks. At large conditioned observer mass, the uncrossed four-point form converges to the independent-probe antipodal product, with an absolute error bound and a first phase correction. Crossed recoupling remains open.

1 Conventions and the reached channel

Set $\ell = 1$, $G_N = 0$, $\mathsf{G} = SO^+(3,1)$. Two scalar BD observers have labels at least $s > 0$. Two distinct free BD matter species have labels $p, q > 0$; local couplings are one. Retain

$$\rho(b) = 2b^2/\pi, \quad \mu(b) = b\tanh(\pi b)/\pi, \quad \sigma(b) = \rho(b)\operatorname{sech}^2 \pi b, \quad h(b) = \sqrt{1+b^2}.$$

For $S = x + y + z$ define

$$D_{xyz} = \cosh(\pi S/2) \prod_{n=x,y,z} \cosh[\pi(S-2n)/2], \tag{1}$$

$$K_{xyz} = \frac{\pi}{8D_{xyz}} \prod_{n=x,y,z} \frac{\sinh \pi n}{n}, \qquad Q_p(a,b) = \frac{\mu(a)\mu(b)}{8D_{abp}}. \tag{2}$$

K is the rigging coefficient of the invariant distribution whose 000 component is one; its coefficient norm is K^{-1}. Emission obeys $|E_{abp}|^2/K_{abp} = \cosh^2(\pi b) Q_p(a,b)$.

The full scalar-pair decomposition contains (J,ν) with $J \in \mathbb{Z}$, not just $J = 0$ [1]. A local emission maps the scalar observer b to observer a plus matter q, so its range has $J = 0$ in this pairing. Changing the pairing generally requires all spins. This reached range is not the whole four-body space.

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2 Clebsch normalization and the physical four-body norm

Normalize the scalar Clebsch transform $C_{aq;b} : \mathcal{H}_a \otimes \mathcal{H}_q \to \mathcal{H}_b$ to have 000 entry one. Its scalar-channel measure is

$$w_{aq}(b) = \rho(b) K_{aqb}, \qquad b > 0. \tag{3}$$

Comparing group averaging with the two-body Schur constant $K_2 = 1/\rho$ gives $K_{aqb} = w_{aq}(b) K_2(b)$. Independently, DS (22), (28), (29) give raw scalar measure $b^2/(4\pi^4)$ on one reflection representative. The boundary kernel in our unitary sphere basis satisfies

$$|B_{000}|^2 = \frac{\pi^4}{D_{aqb}} \prod_{n=a,q,b} \frac{\sinh \pi n}{n}.$$

The measure of $C = B/B_{000}$ is therefore ρK. The reflected full-line factor $1/2$ is already removed. Do not impose the observer cutoff on this mathematical completeness relation.

Let I_{bcp} denote the normalized three-body invariant and put

$$J_b = (C^{\dagger}_{aq;b} \otimes 1_c \otimes 1_p) I_{bcp}.$$

Its all-spherical component is one. Four-body group averaging converges on finite K tests: the product decays as e^{-4r} times a polynomial, and Haar grows as e^{2r} [3, 4]. The scalar part is

$$\eta_4^{(0)} = \int_0^{\infty} \mathrm{d}b \, W |J_b\rangle\langle J_b|, \qquad W(a,b,c;p,q) = \rho(b) K_{abq} K_{bcp}. \tag{4}$$

Other spin channels remain in the full rigging form. The distribution $\int F(b) J_b \, \mathrm{d}b$ has norm $\int |F|^2/W \, \mathrm{d}b$. Compact spectral tests and unitary direct-integral completion justify this statement; no squared delta function is used.

A second creation on a one-matter coefficient $f_1(b,c)$ gives $F(a,b,c) = E_{abq} f_1(b,c)$ with $a, b, c \geq s$, hence

$$\|A_q f_1\|_2^2 = \int_s^{\infty} \mathrm{d}b \, \mathrm{d}c \, \frac{|f_1(b,c)|^2}{K_{bcp}} \mathcal{S}_q(b), \tag{5}$$

$$\mathcal{S}_q(b) = \int_s^{\infty} R_q(a|b) \, \mathrm{d}a, \qquad R_p(a|b) = Q_p(a,b)/\sigma(b). \tag{6}$$

This fiber multiplication map is closed. Compact external mass supports give a dense domain in $\mathcal{H}_1$; its adjoint annihilates the orthogonal complement of the closure of the reached range.

For vacuum coefficient $f(c)$, two creations give $f(c) E_{bcp} E_{abq}$. In the linear-gluing state $f(c) = \operatorname{sech} \pi c$, the squared-norm density is

$$\mathcal{W}_{pq}(a,b,c) = Q_q(a,b) Q_p(b,c)/\sigma(b). \tag{7}$$

The chord-kernel structure has a direct dS_2 predecessor, KL (6.23)–(6.25) [2]; our dS_3 factors follow from the fixed local model.

3 Domains and a proper-time half-period shift

The unshifted rate obeys

$$\mathcal{S}_q(b) \le b/(2\pi) + c_s, \qquad c_s = 1/(2\pi^2) + 1/(4\pi^3 s).$$

From $R_p(b|c) \le be^{-2\pi(b-c)_+}/(\pi c)$, with $\beta = 2\pi$,

$$\int_s^\infty bR_p(b|c)\,\mathrm{d}b \le \frac{c^2}{3\pi} + \frac{c}{\pi\beta} + \frac{2}{\pi\beta^2} + \frac{2}{\pi\beta^3 s}. \tag{8}$$

Thus a finite second mass moment of $\rho(c)|f(c)|^2\,\mathrm{d}c$ is sufficient for A_qA_pf; the linear-gluing state meets it. This does not establish a common full-Fock core. Its ordered two-particle overlap is

$$Z^{-1}\int_{a,b,c\ge s} \mathrm{d}a\,\mathrm{d}b\,\mathrm{d}c\,\mathcal{W}_{pq}e^{-it_p[h(b)-h(c)]-it_q[h(a)-h(b)]}.$$

Unshifted conditional rates have a broad downward-energy region at large mass. Their coincidence norms are not the finite antipodal benchmark.

On compact mass tests define

$$\widetilde{A}_p = e^{\pi H_{\mathrm{out}}/2}A_pe^{-\pi H_{\mathrm{in}}/2}.$$

For $A_p(t) = e^{-iH_{\mathrm{out}}t}A_pe^{iH_{\mathrm{in}}t}$ this is $t = i\pi/2$. Its conditional norm kernel has the exact factorization

$$\widetilde{R}_p(a|b) = e^{\pi[h(a)-h(b)]}R_p(a|b) = \frac{g(a)}{g(b)}\widehat{R}_p(a,b), \tag{9}$$

$$g(x) = \frac{xe^{\pi[h(x)-x]}}{1+e^{-2\pi x}}, \tag{10}$$

$$\widehat{R}_p(a,b) = \frac{\cosh\pi(a+b) - \cosh\pi(a-b)}{8\pi[\cosh\pi(a+b)+\cosh\pi p][\cosh\pi(a-b)+\cosh\pi p]}. \tag{11}$$

This does not identify the label with proper energy or prove geometric antipodality at finite mass. Set

$$k_p(u) = \frac{1}{8\pi(\cosh\pi u + \cosh\pi p)}, \qquad B_s = 2e^{\pi(\sqrt{1+s^2}-s)}.$$

Then $0 < \widehat{R}_p(a,b) \le k_p(a-b)$ and $g(a)/g(b) \le B_s(1+|a-b|/s)$. Using $k_p(u) \le e^{-\pi|u|}/(4\pi)$ gives

$$\sup_{b\ge s}\int_s^\infty \widetilde{R}_p(a|b)\,\mathrm{d}a \le B_s\left(\frac{1}{2\pi^2} + \frac{1}{2s\pi^3}\right). \tag{12}$$

Each shifted creation block therefore extends boundedly on the sectors treated here. The exponentials in its definition need not act separately on arbitrary states.

4 A conditioned four-point form and its limit

Condition on initial observer mass $c = M$. More precisely, normalized smooth vacuum packets whose mass probability tends to $\delta(c - M)$ give the continuous fiber form

$$F_M(t_p, t_q) = \int_{a,b\geq s} \mathrm{d}a\, \mathrm{d}b\, \widetilde{R}_q(a|b)\widetilde{R}_p(b|M)e^{-it_p[h(b)-h(M)]-it_q[h(a)-h(b)]}. \tag{13}$$

The creation-state overlap fixes the ordering. This is an uncrossed form, not the unconditioned HH expectation or a crossed OTOC. Write $u = b - M$, $v = a - b$. The g ratios telescope to $g(M+u+v)/g(M)$, which tends to one. The balanced factors tend to $k_p(u)k_q(v)$ and the proper energy differences tend to u, v. An integrable dominating function is

$$B_s\left(1 + \frac{|u| + |v|}{s}\right) k_p(u)k_q(v).$$

With s fixed, the cutoff region vanishes pointwise. Uniformly on compact real time sets,

$$\boxed{F_M(t_p, t_q) \longrightarrow L_p(t_p)L_q(t_q), \qquad L_p(t) = \frac{\sin pt}{4\pi \sinh \pi p \, \sinh t}.} \tag{14}$$

The continuous value is $L_p(0) = p/(4\pi \sinh \pi p)$. Thus

$$F_M(0, 0) \longrightarrow \frac{pq}{16\pi^2 \sinh \pi p \, \sinh \pi q}.$$

This recovers the independent-probe antipodal benchmark. The half-period continuation is part of the specified observable; a moving cutoff, identical species or removal of the shift is a different limit.

For an explicit bound let $M \geq 2s$, $T = \max(|t_p|, |t_q|)$ and

$$d_M = \frac{2}{M} + \frac{2\pi}{M^2} + 2\pi e^{-\pi M}, \quad C_M = e^{Md_M/2}, \quad \epsilon_M = 2(2 + \cosh \pi p + \cosh \pi q)e^{-\pi M}.$$

Define

$$\mathcal{T}_s(R) = \frac{e^{-\pi R}}{4\pi^2}\left[(1 + B_s)\left(\frac{R}{\pi} + \frac{1}{\pi^2}\right) + \frac{B_s}{s}\left(\frac{R^2}{\pi} + \frac{2R}{\pi^2} + \frac{2}{\pi^3}\right)\right]. \tag{15}$$

Then

$$|F_M - L_pL_q| \leq \frac{C_M(d_M + 2T/M^2)}{2\pi^5} + \epsilon_M L_p(0)L_q(0) + \mathcal{T}_s(M/2). \tag{16}$$

This is a conservative absolute estimate, not a small finite-mass percentage guarantee. Relative estimates divide by $|L_pL_q|$ only away from its zeros.

5 Error control and the first finite-mass phase

Split the difference plane at $r = |u| + |v| = M/2$. Inside, all masses are at least $M/2$:

$$|(\log g)'| \le d_M, \qquad |g(M+u+v)/g(M) - 1| \le C_M d_M r.$$

The balanced-factor deficits sum to at most ϵ_M. Since $1 - h'(x) \le 1/(2x^2)$, the phase error is at most $2Tr/M^2$. Use $\int r k_p(u) k_q(v)\,\mathrm{d}u\,\mathrm{d}v \le 1/(2\pi^5)$. Outside, actual plus limiting kernels are bounded by $[1 + B_s(1 + r/s)]e^{-\pi r}/(16\pi^2)$; four-quadrant integration gives $\mathcal{T}_s$. This proves the previous $O_T(M^{-1})$ estimate.

There is also a sharper asymptotic:

$$F_M = L_p L_q + \frac{i}{M}\left[L_p'(t_p)L_q(t_q) + L_p(t_p)L_q'(t_q)\right] + O_T(M^{-2}). \tag{17}$$

On the inner diamond, $g(M+u+v)/g(M) = 1+(u+v)/M+O((r+r^2)/M^2)$; the balanced deficits are exponential, and the proper-energy phase error is $O_T(r/M^2)$. All polynomial moments of $k_p k_q$ are finite; the outer tail, including the subtracted linear term, is exponentially small. Finally $\int u k_p(u) e^{-itu}\,\mathrm{d}u = iL_p'(t)$. The first finite-mass correction is imaginary. The saved second-order numerical expression is exploratory and is not used as a proved remainder or a pass criterion.

6 Independent group-normalization checks

The main run has 56 checks; a separate angular run has 10. All pass. DS sphere normalization is compared to ρK in three samples. Six spherical-product reconstructions test $\phi_a(r)\phi_q(r) = \int_0^\infty w_{aq}(b)\phi_b(r)\,\mathrm{d}b$, including $r = 0$.

Three four-body samples compare direct radial and spectral integrals with

$$\int_0^\infty \sinh^2 r \prod_{j=1}^{4} \phi_{n_j}(r)\,\mathrm{d}r = \frac{1}{16\prod_j n_j} \sum_{\varepsilon_j = \pm 1} \left(\prod_j \varepsilon_j\right) f\left(\sum_j \varepsilon_j n_j\right), \tag{18}$$

where $f(k) = 1 - (\pi k/2)\coth(\pi k/2)$ and $f(0) = 0$. Expand four sines and use $\int(\cos kr - 1)/\sinh^2 r\,\mathrm{d}r = f(k)$. The sample $(1,1,1,1)$ checks zero frequencies. The $l = 1$ pair singlet has overlap

$$I_{aqb}(S_1) = \frac{\sqrt{3}(1 + a^2 + q^2 - b^2)}{2\sqrt{(1+a^2)(1+q^2)}}.$$

Six local-vertex checks and two independent non-spherical group integrals test this component. Radial cutoff 40 and spectral cutoff 18 have analytic outer tails. This does not numerically test every spinning channel.

7 Numerical benchmark and continuation

The proper-time run uses $s = .2$, $p = .6$, $q = 1.2$, box $|u|, |v| \le 14$, Gauss orders 64/96, and the exact moving cutoff. The outside-box error is at most $\mathcal{T}_{.2}(14) < 1.75\,10^{-17}$. The interior quadrature is empirical, not interval-certified.

M	$F_M(0,0)$	Relative difference from limit
4	.00006571136262	$5.0433\,10^{-3}$
8	.00006540451475	$3.5012\,10^{-4}$
16	.00006538310306	$2.2634\,10^{-5}$
32	.00006538171794	$1.4491\,10^{-6}$

The limit is .00006538162320. At $(t_p, t_q) = (.5, 1)$, the complex relative differences are 16.7%, 10.8%, 6.04% and 3.18%. Subtracting the first phase term leaves absolute residuals

$$4.44\,10^{-6}, \quad 1.11\,10^{-6}, \quad 2.79\,10^{-7}, \quad 6.98\,10^{-8}.$$

Small zero-time errors do not certify the full finite-time correlator.

The crossed problem is next: transform between the (a, q) and (a, p) pairings with the principal spin channels, reflection representative, and exact leg phases. A scalar range in one pairing cannot be treated as a complete scalar range in another. The prior normalization-invariant mismatch with the candidate recoil graph is not removed by this norm identity. Identical species require Bose interference between pairings. A common full-Fock domain, composite gluing, finite-κ sewing and finite-G_N dynamics remain open; no full holographic dual is claimed.

Implementation and two validation drivers are in `code/`. Raw values, parameters, dependency versions and source hashes are in `results/stage4/ds3-two-matter/`; the Chinese derivation is `notes/stage4-ds3-two-matter.md`. Group and vertex checks use 50-digit arithmetic; time integrals use NumPy double precision. The first-correction remainder above is an analytic asymptotic statement, not an interval-certified numerical error.

References

[1] S. E. Derkachov and V. P. Spiridonov, *On the 6j-symbols for SL(2,C) group*, arXiv:1711.07073v2, (17), (22), (28)–(30), and Sec. 4.

[2] D. K. Kolchmeyer and H. Liu, *Chaos and the Emergence of the Cosmological Horizon*, arXiv:2411.08090v1, Secs. 6.2–6.3, especially (6.23)–(6.25).

[3] D. Marolf and I. A. Morrison, *Group Averaging for de Sitter free fields*, arXiv:0810.5163v1.

[4] D. Marolf and Z. Yu, *On the Positivity of de Sitter Group Averaging*, arXiv:2610.24073v1, Secs. 5.1–5.2.